LOAN DEFAULT AND CREDIT RISK ANALYSIS

Project Report

1. Project Objective

The objective of this project is to examine loan application and borrower information to understand patterns associated with recorded loan default. The analysis compares default outcomes across loan purpose, employment experience, home ownership, credit score, income, loan amount, debt-to-income ratio, and other available fields.

The results are descriptive. They summarize patterns in the supplied dataset and do not constitute a validated credit scoring model or a recommendation to approve or reject any individual applicant.

2. Dataset Description

The cleaned dataset contains 1,000 loan records and 17 fields. These include a loan identifier, borrower age, annual income, employment years, loan amount and term, credit score, existing debt, debt-to-income ratio, open accounts, delinquencies, home ownership, loan purpose, loan grade, interest rate, monthly payment, and loan status.

The outcome field is **Loan _Status**, with the categories Default and Paid. The analysis derives a binary default flag from this field. The supplied dataset is a fixed project sample; findings describe this sample only.

3. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI Desktop, Power Query, and DAX.

4. Data Cleaning

The Python workflow loads the source CSV, checks field types and summary statistics, checks missing values and duplicate records, and standardizes loan status text before deriving the default flag. The cleaning routine removes duplicate rows and is set up to fill missing numeric values with the median and missing categorical values with the mode when needed.

For the supplied 1,000-row dataset, the cleaned file contains no missing values and no duplicate rows. The final cleaned dataset is saved as **loan_default_credit_risk_cleaned.csv**. Potential outliers are counted using the interquartile-range method for review; they are not automatically removed.

5. Data Analysis

The analysis calculates total loans, defaulted and paid loans, overall default rate, and averages for annual income, credit score, loan amount, interest rate, and debt-to-income ratio. It also compares default rates by loan purpose, employment experience group, home ownership, loan grade, credit-score group, and income group.

The supplied sample contains 1,000 loans: 932 are recorded as Default and 68 as Paid, giving an observed default rate of 93.20%. Mean credit score is 685.59, mean annual income is 51,733.81, mean loan amount is 211,514.41, mean interest rate is 7.36, and mean DTI value is 4.33 in the dataset units.

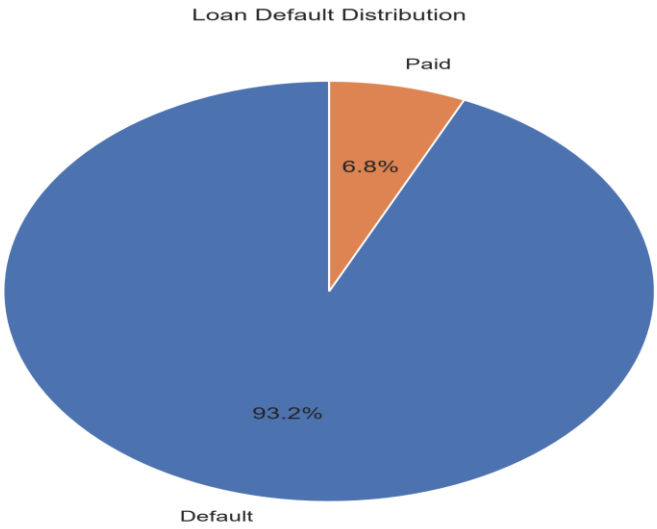
The outcome is highly imbalanced: 93.2% of records are defaults. Any interpretation or later predictive modeling should first verify the source data and class balance. The DTI

scale should also be confirmed before interpreting its magnitude.

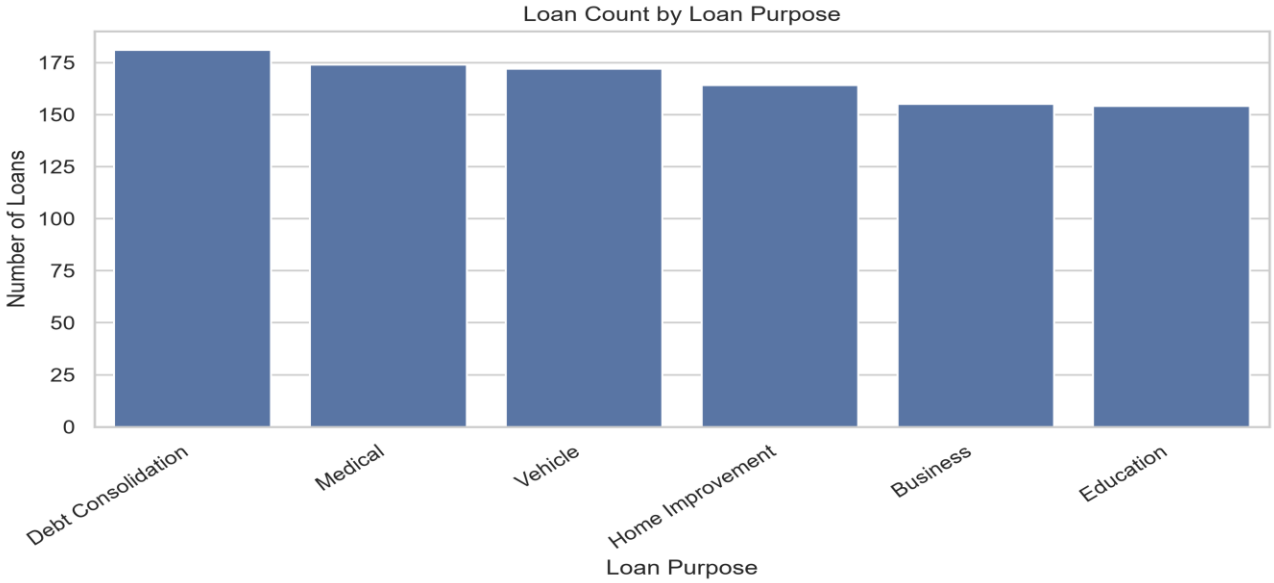
6. Python Visualizations

The project includes the following visualizations, saved as PNG files in the visualizations folder:

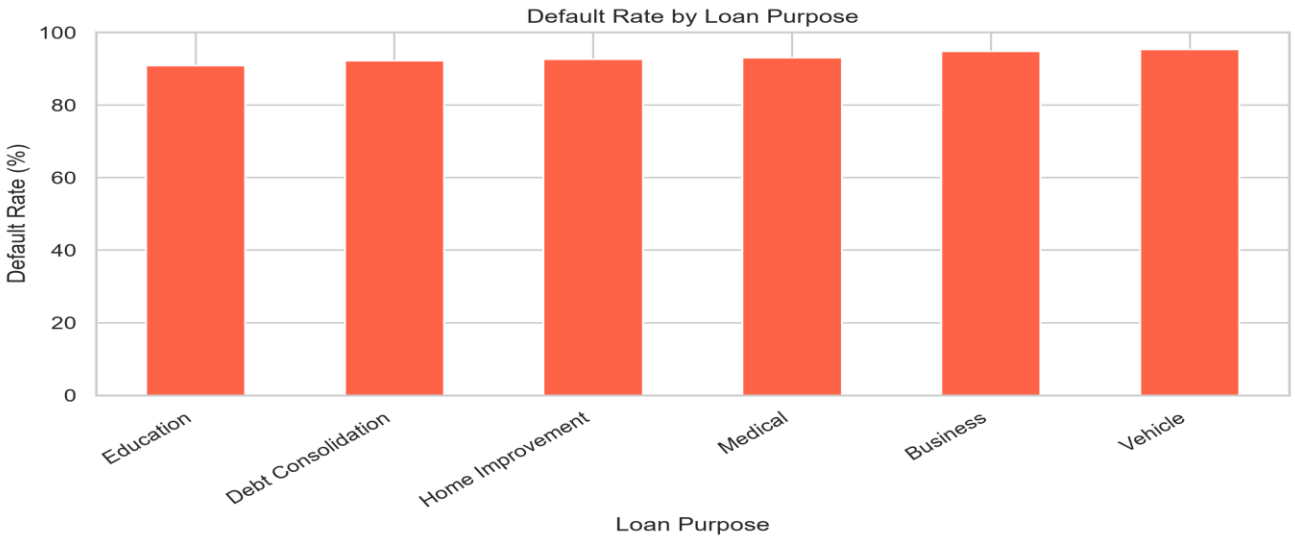
• **Loan default distribution**



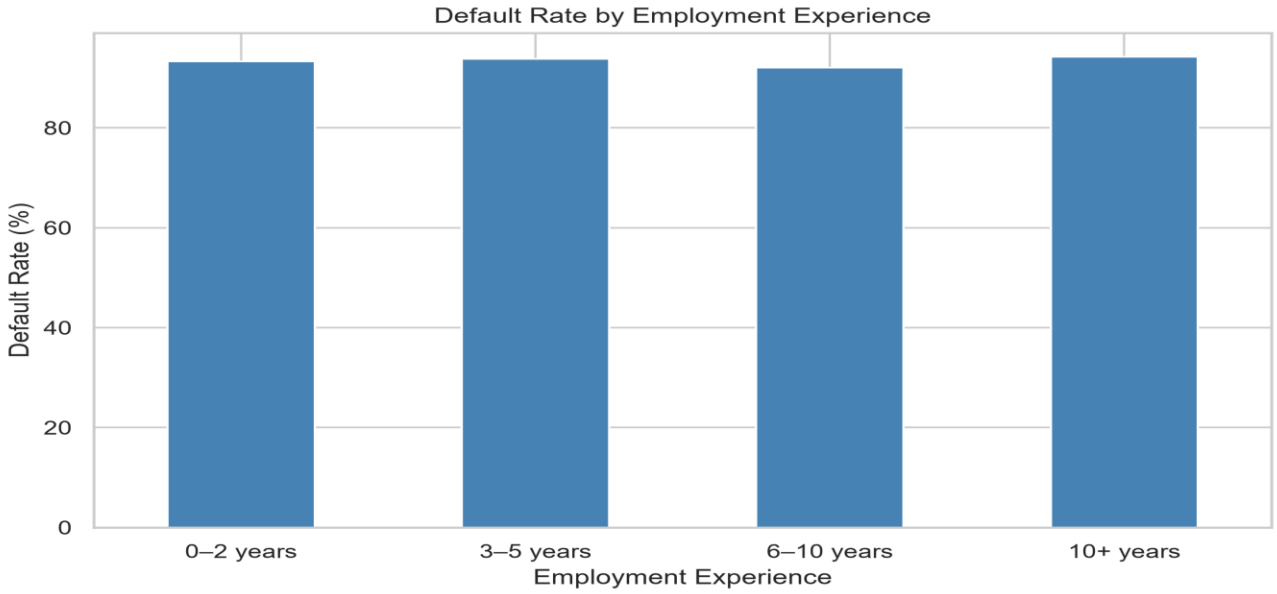
• **Loan count by purpose**



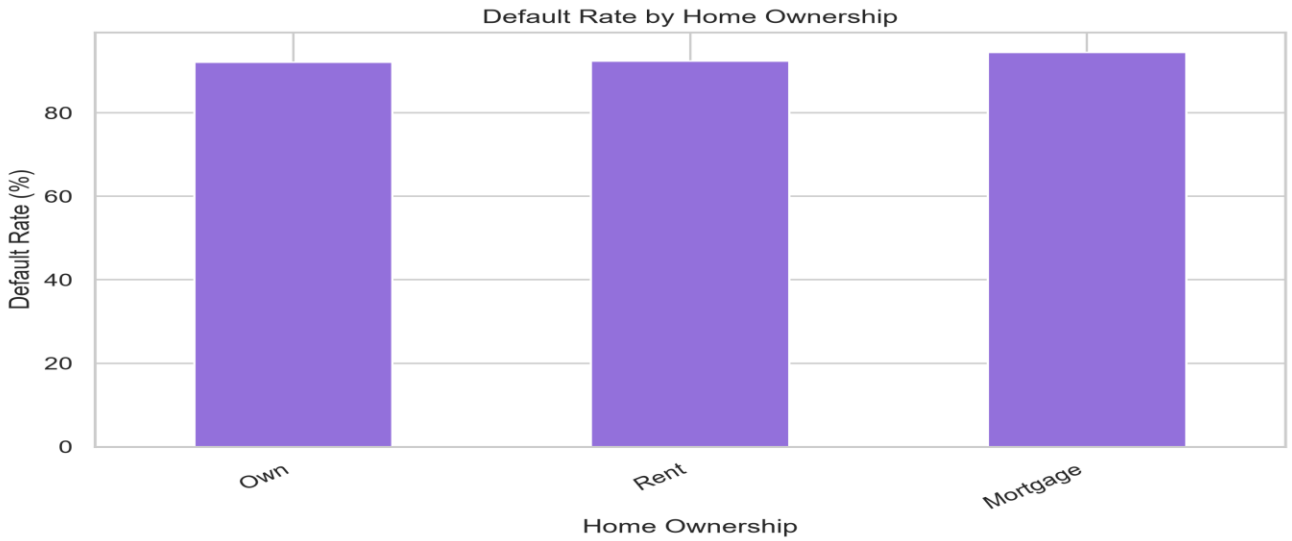
• **Default rate by purpose**



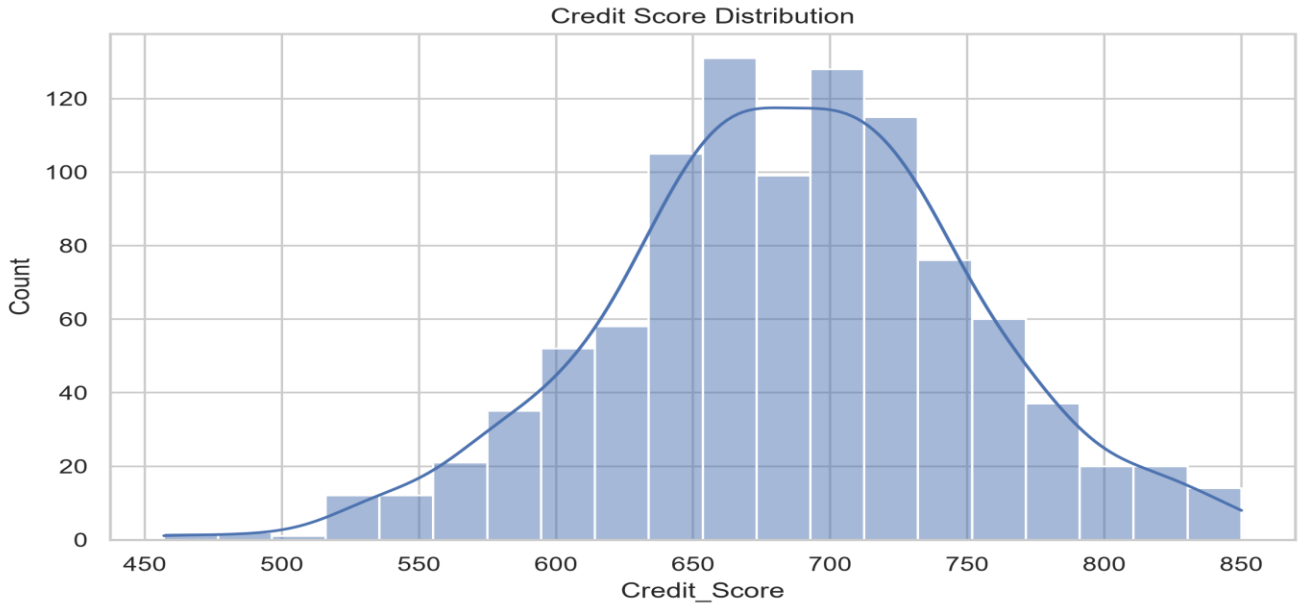
- **Default rate by employment experience**



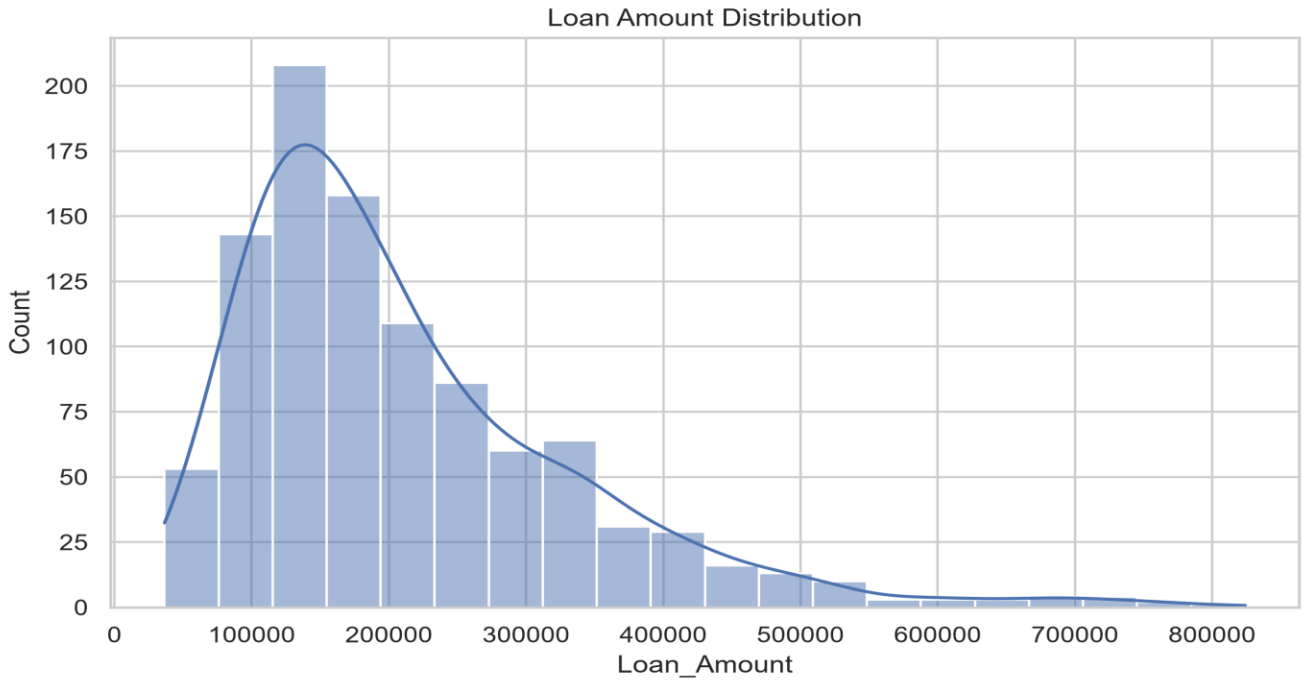
- **Default rate by home ownership**



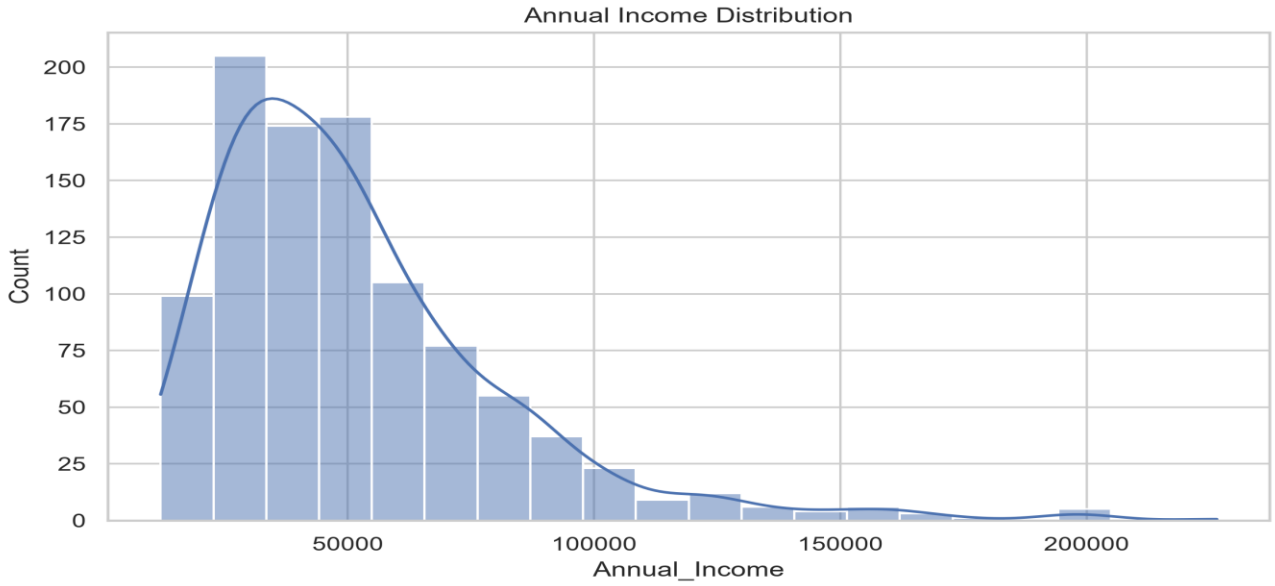
- **Credit score distribution**



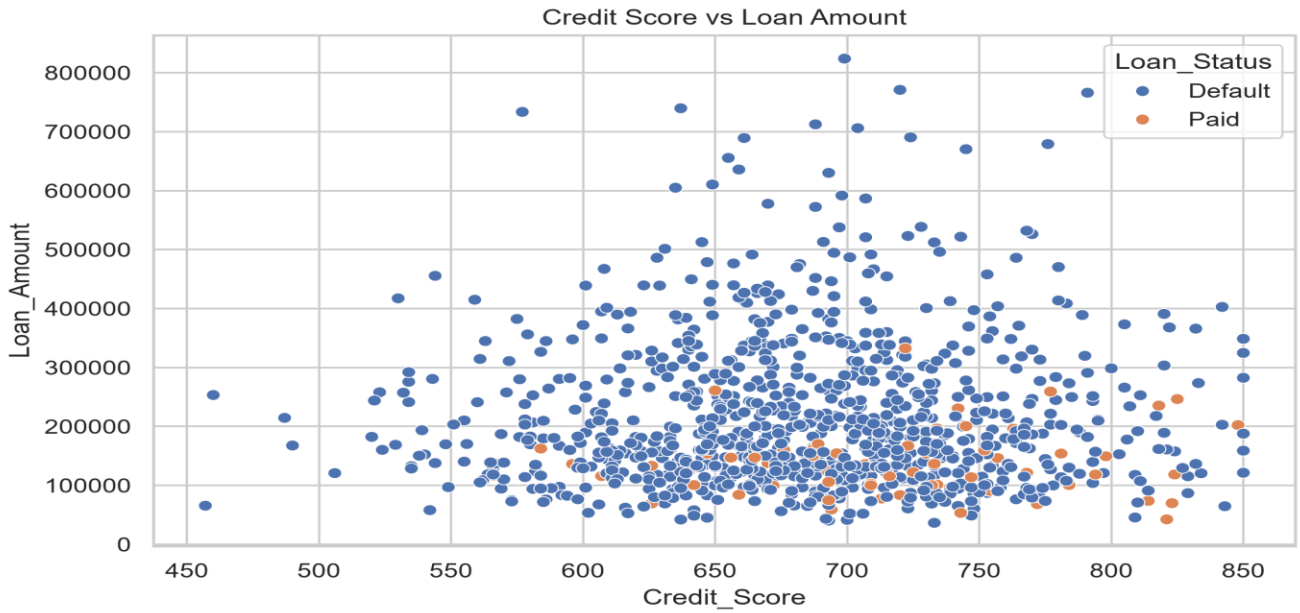
- **Loan amount distribution**



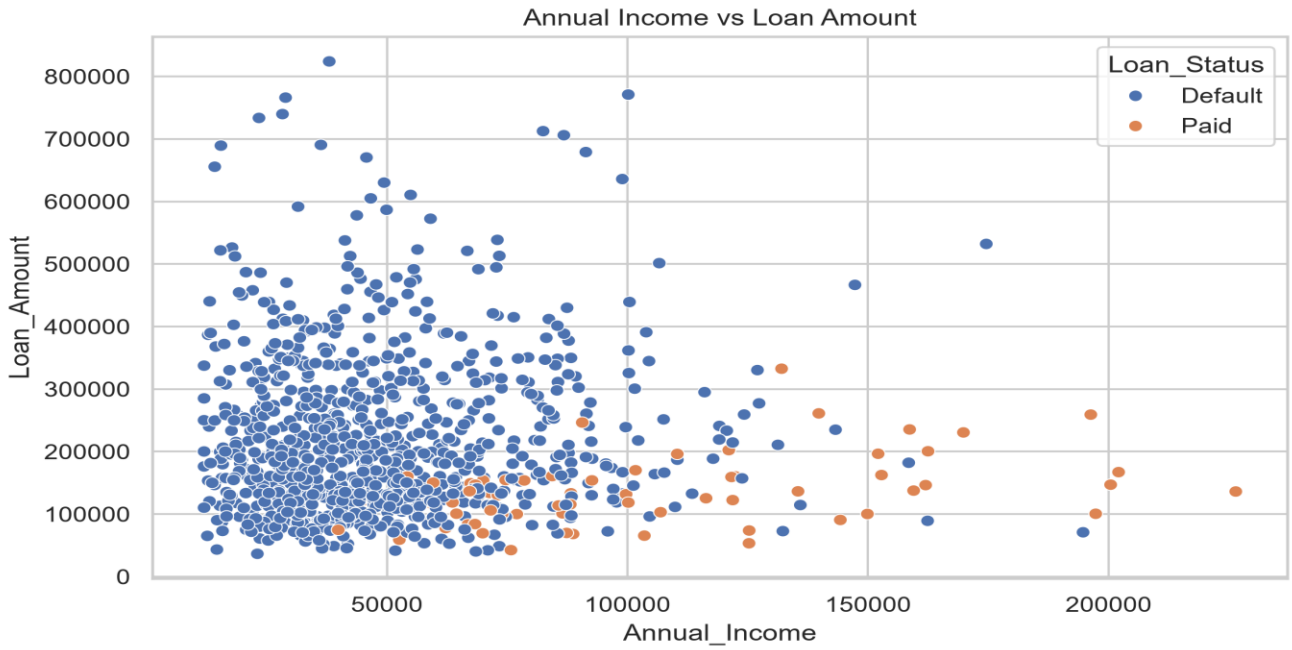
- **Annual income distribution**



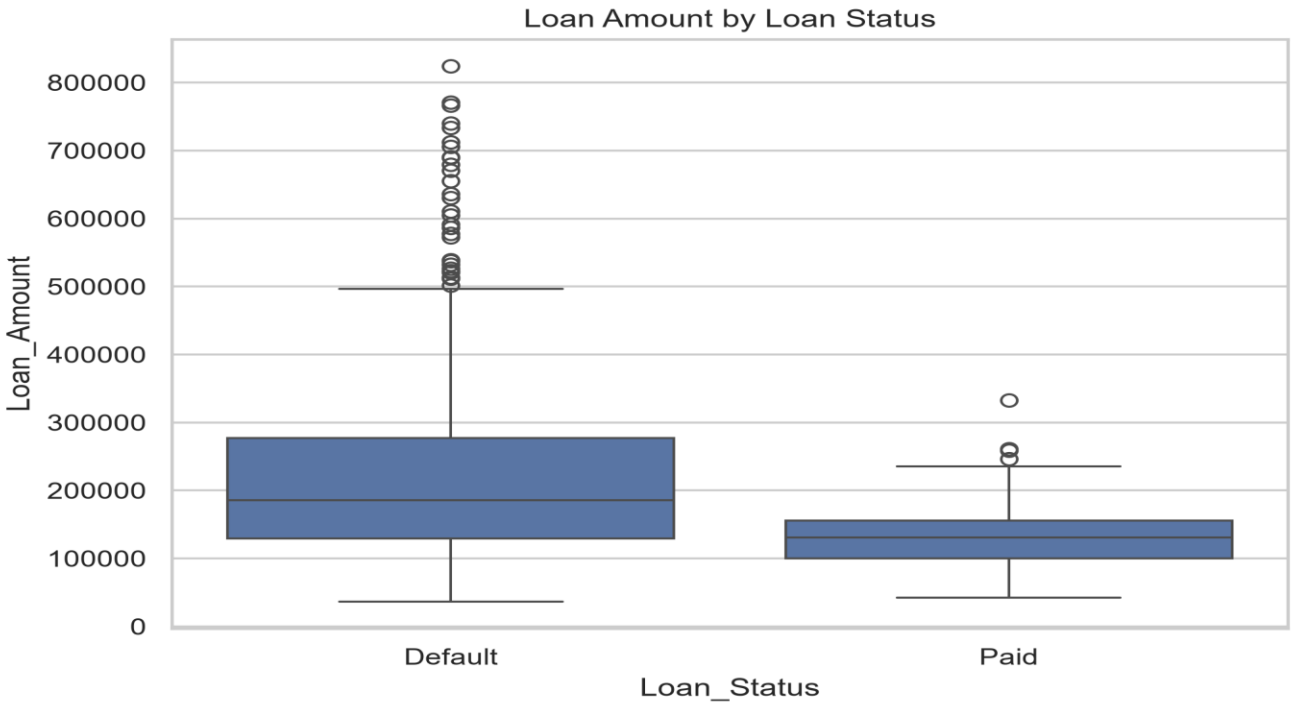
- **Credit score versus loan amount**



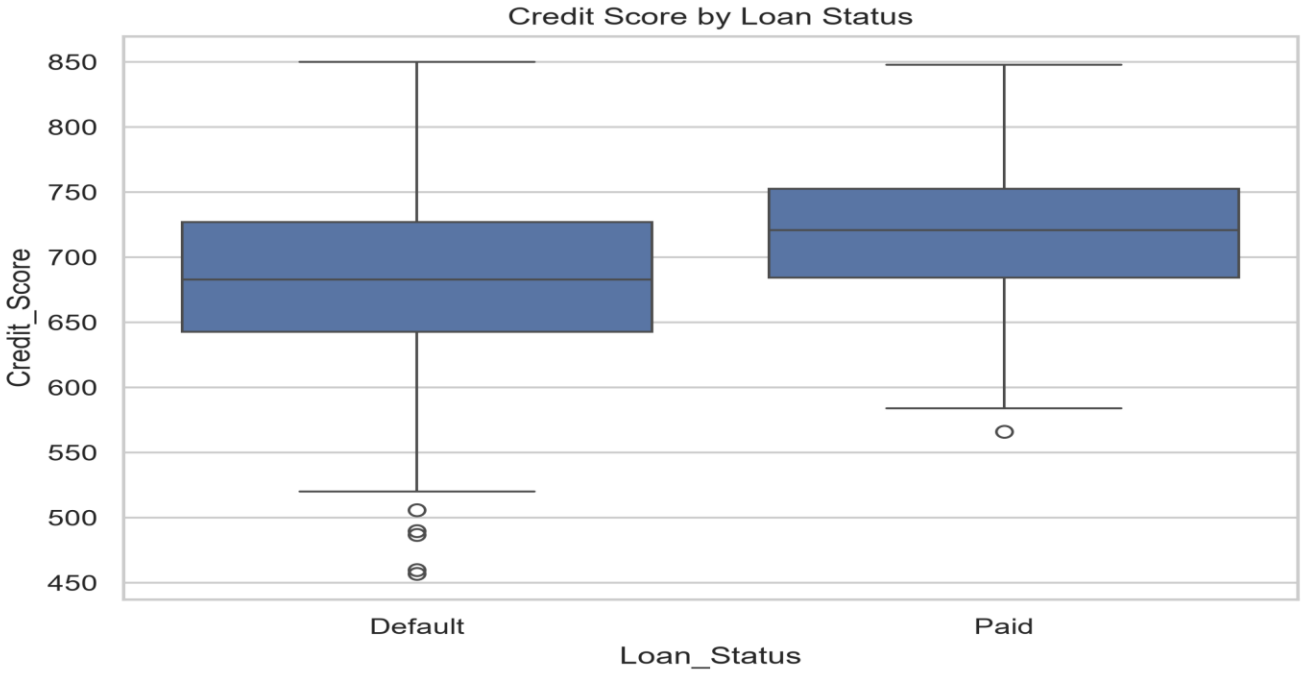
- **Income versus loan amount**



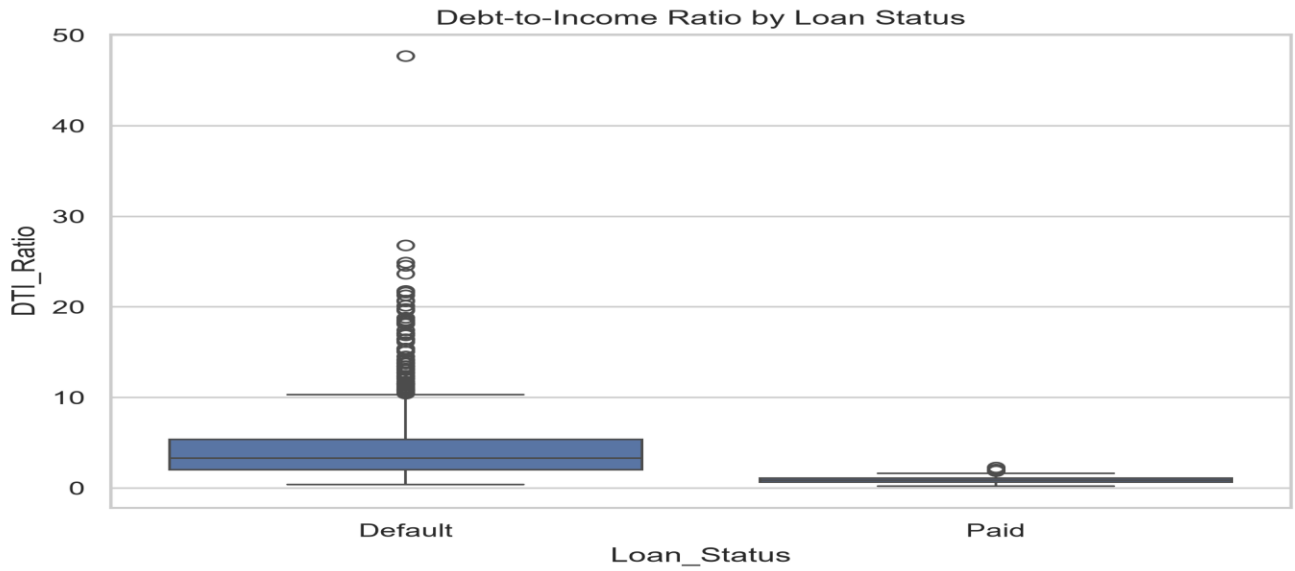
- **Loan amount by repayment status**



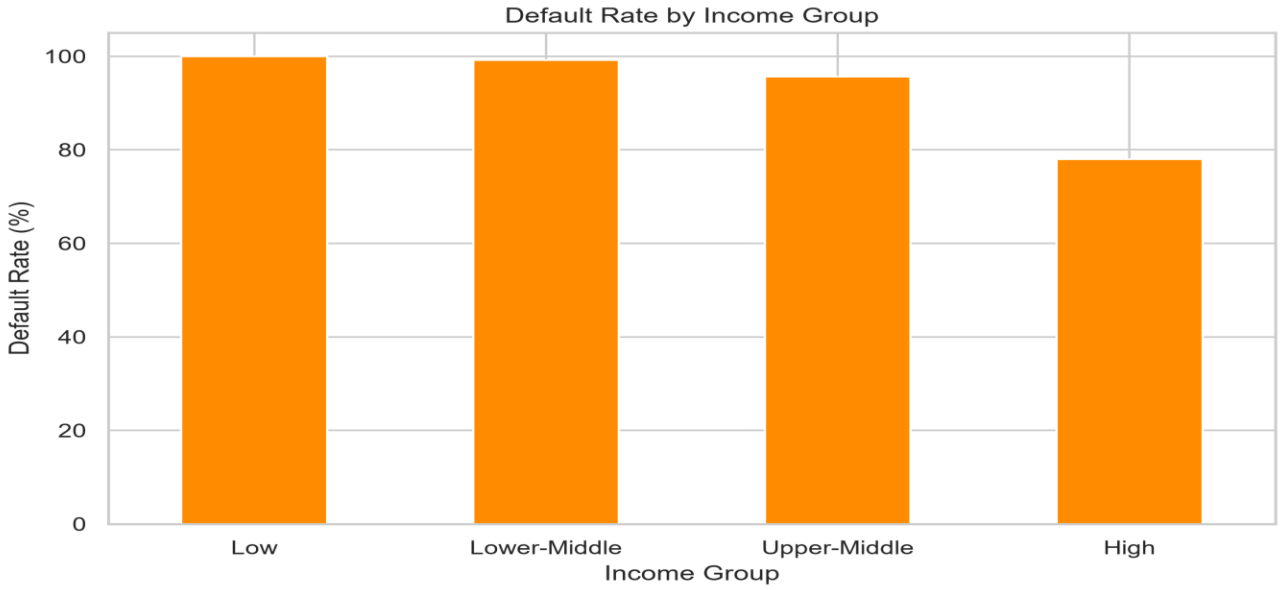
- **Credit score by repayment status**



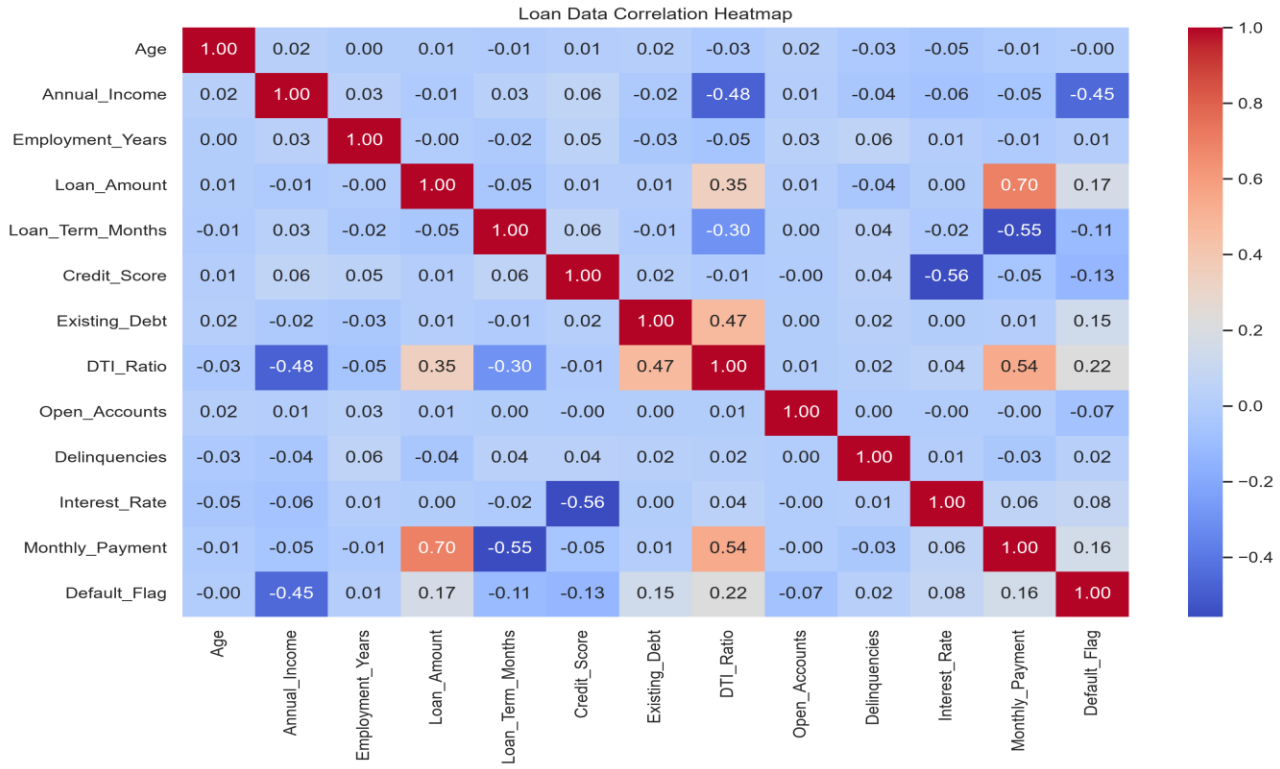
- **DTI ratio by repayment status**



- **Default rate by income group**



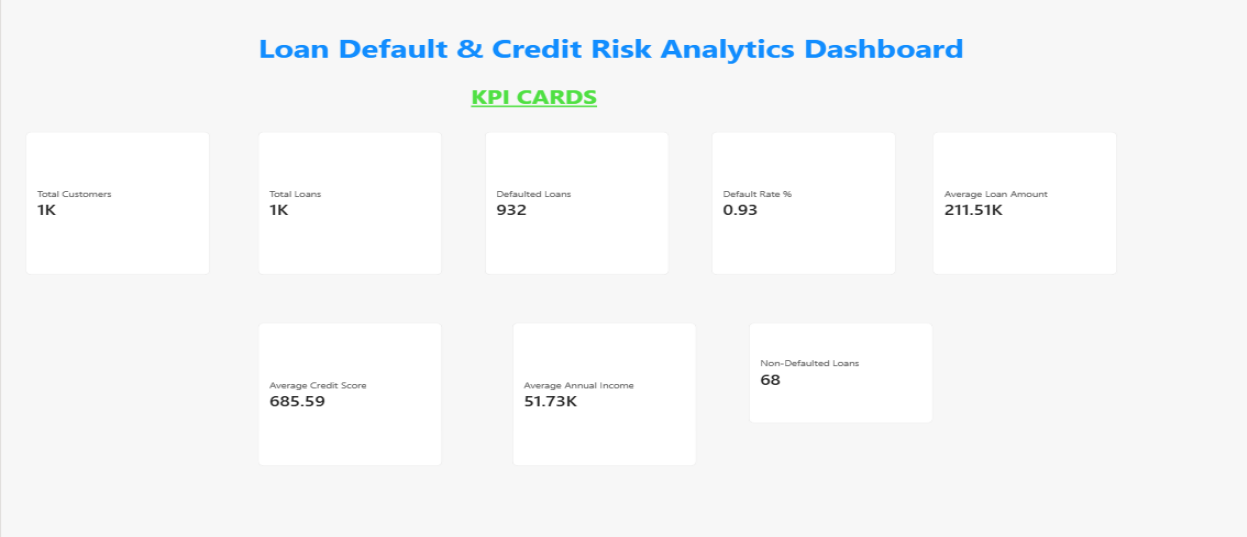
- **Correlation heatmap**



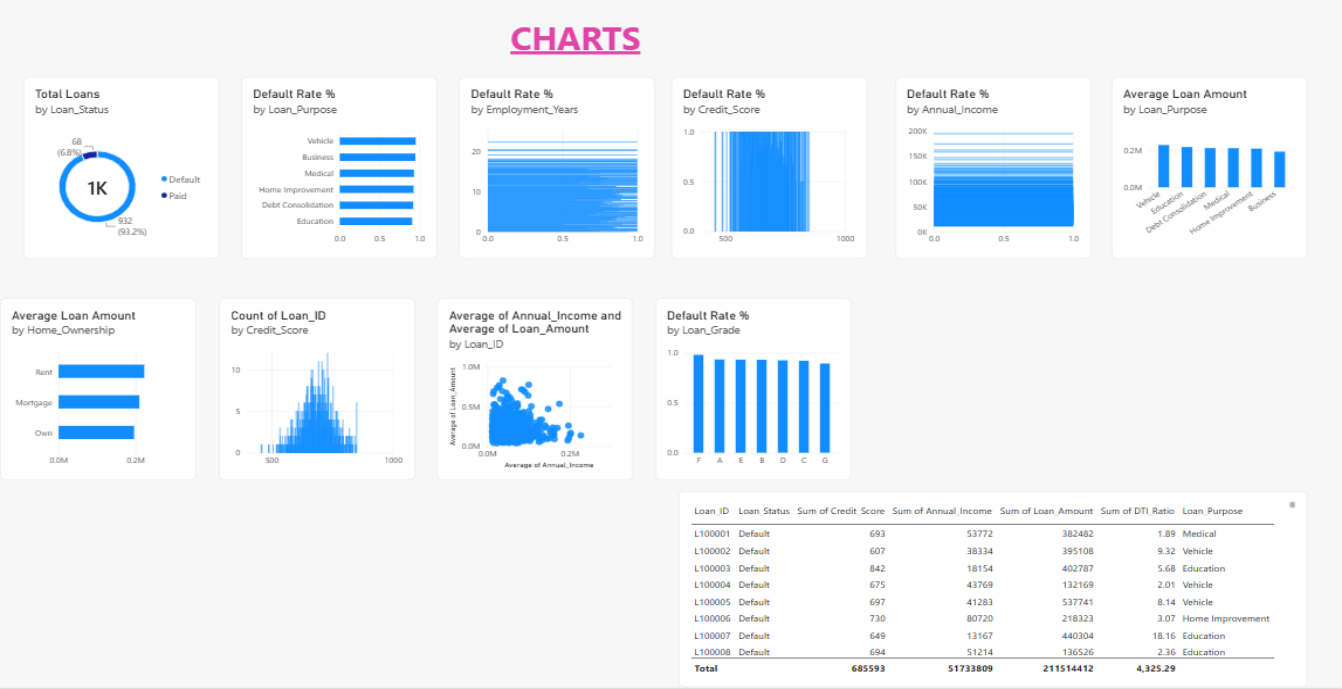
7. Power BI Dashboard

The Power BI project file is supplied as **loan_default_risk_dashboard.pbix**. It is intended to present the loan portfolio and default patterns through summary indicators and category comparisons. Add the dashboard screenshots to this report before final submission, as the supplied project folder contains the PBIX file but no exported dashboard screenshots.

KPI cards

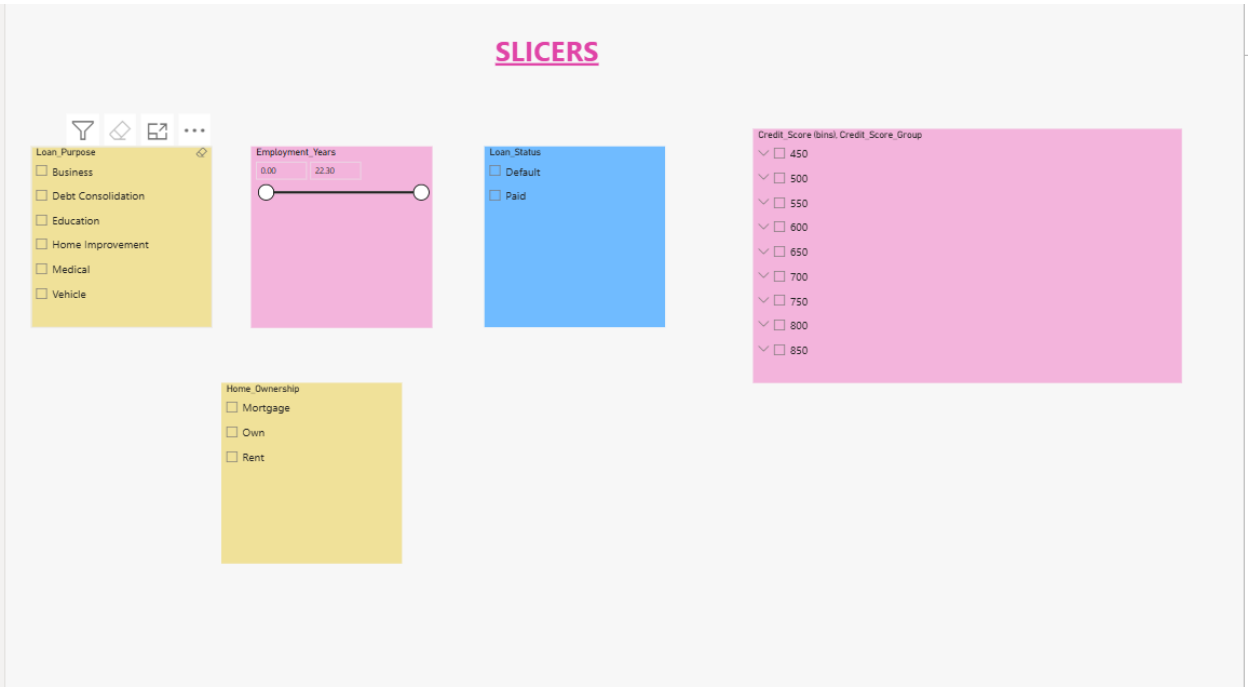


CHARTS



Loan_ID	Loan_Status	Sum of Credit_Score	Sum of Annual_Income	Sum of Loan_Amount	Sum of DTI_Ratio	Loan_Purpose
L100001	Default	693	53772	382482	1.89	Medical
L100002	Default	607	38334	395108	9.32	Vehicle
L100003	Default	842	18154	402787	5.68	Education
L100004	Default	675	43769	132169	2.01	Vehicle
L100005	Default	697	41283	537741	8.14	Vehicle
L100006	Default	730	80720	218323	3.07	Home Improvement
L100007	Default	649	13167	440304	18.16	Education
L100008	Default	694	51214	136526	2.36	Education
Total		**685593**	**51733809**	**211514412**	**4,325.29**	

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8. Key Insights

In the supplied sample, 932 of 1,000 loans are labeled Default (93.20%), while 68 are labeled Paid (6.80%). This strong imbalance is a central limitation when reading category comparisons.
Average credit score is lower among default records (683.26) than paid records (717.57). Average annual income is 48,072.04 for default records and 101,921.63 for paid records.
Average loan amount is higher among default records (217,098.62) than paid records (134,977.87).
Average interest rate is 7.41 for default records and 6.75 for paid records.

By loan purpose, Vehicle has the highest observed default rate at 95.35% (172 records), followed by Business at 94.84% (155 records). Education has the lowest observed rate at 90.91% (154 records). By home ownership, Mortgage records show a 94.55% default rate (385 records). By loan grade, Grade F has the highest observed rate at 98.18% (55 records); this subgroup is comparatively small.Loan Default and Credit Risk Analysis

The DTI values average 4.57 for default records and 0.94 for paid records. Since the dataset does not document the DTI scale, this difference should be reported as stored and interpreted only after confirming the units.

9. Conclusion

This project summarizes loan repayment status and borrower characteristics in the supplied dataset. The sample shows associations between recorded default status and credit score, income, loan amount, loan purpose, home ownership, and loan grade. These are descriptive associations and do not establish that any factor causes default.

The unusually high default share and undocumented DTI scale should be checked against the source before drawing operational conclusions. The analysis can be extended by validating the data, addressing class imbalance, documenting metric definitions, and evaluating a properly validated risk model if predictive scoring is required.

10. Project Files

The project package includes the source and cleaned loan datasets, Python analysis script, saved visualization images, and Power BI dashboard file. The report text is provided without embedded chart images so that the project owner can insert the selected Python and Power BI screenshots before submission.